

CreditCardPayment.java - Code Structure & Links

1. CLASS DEFINITION

```
public class CreditCardPayment extends Payment
```

Explanation: This is a Concrete Subclass. The "extends" keyword links it to the Payment base class. It inherits all base fields (amount, currency, status) but adds its own specific logic.

2. CLASS VARIABLES (FIELDS)

```
private String cardNumber;  
private String cardHolderName;  
private String expiry;  
private String cvv;
```

Explanation: These are "private" variables. They only exist inside a Credit Card payment. The PaymentProcessor does not know they exist, and the base Payment class does not know they exist. This completely isolates the credit card data.

3. CONSTRUCTOR

```
public CreditCardPayment(...)
```

Explanation: Inside the constructor, it immediately calls super(amount, currency). This passes the generic data UP to the parent Payment class so the parent can handle the ID generation and timestamp. Then, it saves the specific credit card details into its private variables.

4. METHODS & ALGORITHMS

```
public String maskedNumber()
```

Explanation: A utility method that strips spaces from the card number and returns only the last 4 digits (e.g., ••• 1234) for security logging.

```
@Override  
public boolean process(BiConsumer<String, String> log)
```

Explanation: This fulfills the contract demanded by the parent class.

Algorithm Flow:

1. It logs the masked card details.
2. It runs a validation check: ensures the card has 12+ digits and uses Regex (`^\d{3}$`) to verify the CVV is 3 digits. If this fails, it sets status="FAILED" and stops.
3. It simulates network authorization (declining artificially if amount > \$500,000). If successful, it updates status="SUCCESS" and returns true.